

Sahil Khan

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Software engineer focused on backend development and applied AI, with merged open-source contributions.

PROJECTS

AmazonHelp Support Agent

Sep 2026

Python, Gemini API, sentence-transformers, scikit-learn

- Built intent classification, semantic retrieval, and grounded support-reply generation.
- Added TF-IDF fallback, deterministic support links, and human-escalation rules.
- Implemented human/LLM scoring tools and weighted-kappa agreement analysis.

CodebaseToSpec

Aug 2026

Python, AST, MongoDB, Pydantic, Gemini API

- Indexed Python files, symbols, and imports in MongoDB using AST parsing.
- Built a bounded tool-calling loop generating structured plans and execution traces.
- Verified file/symbol references and implemented bounded re-planning on errors.

SIS Migration Engine

Jul 2026

TypeScript, Node.js, GraphQL, PostgreSQL, BullMQ, Redis

- Built staged CSV imports with schema validation, reference checks, and duplicate detection.
- Implemented GraphQL review workflows for editing records and resolving conflicts.
- Queued approved records with BullMQ and isolated failures using PostgreSQL transactions.

StoreIt

Apr 2025 – Jun 2025

Next.js, TypeScript, Appwrite

- Built file upload and metadata-storage workflows using Next.js and Appwrite.
- Implemented file search, renaming, sharing, and deletion.
- Added failure cleanup to remove uploaded files when metadata creation failed.

CarePulse

Oct 2024 – Nov 2024

Next.js, TypeScript, Appwrite, Zod, Twilio

- Built patient registration workflows with Zod validation and Appwrite data storage.
- Implemented appointment scheduling and cancellation workflows.
- Integrated Twilio SMS notifications for appointment updates.

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL

Backend & Data: Node.js, Express, GraphQL, PostgreSQL, MongoDB, Redis, BullMQ, Knex

AI & Validation: Gemini API, sentence-transformers, scikit-learn, TF-IDF, Pydantic, Zod

Web & Tools: React, Next.js, Git, Docker

ACHIEVEMENTS & OPEN SOURCE

- **Turborepo** — **PR #14366, merged Oct 5, 2026:** Fixed empty task graphs in affected/tag-filtered package listings; added Rust regression tests.
- **Turborepo** — **PR #10579, merged Jun 19, 2025:** Restored shared-component styling with the missing Tailwind import.
- Solved **700+** problems on [LeetCode](#).

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University

Jul 2019 – Jul 2023

Bachelor of Technology, Computer Science and Engineering

Kanpur, India

International Indian School

2016 – 2019

High School (2018–2019) & Secondary School (2016–2017)

Dammam, Saudi Arabia